

Damage Mechanics in FRACMAT

A worked derivation of the scalar CDM model,
the modified von Mises equivalent strain, and the Newton–Raphson solver

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Abstract

This manual derives the constitutive model behind FRACMAT, step by step, with nothing swept under the rug. It covers the scalar isotropic damage law, the modified von Mises equivalent strain of de Vree and co-workers, exponential softening, crack band regularization through Oliver’s characteristic length, and the modified Newton–Raphson scheme the code actually solves with. Where a formula looks like it was pulled from thin air, we check it against a uniaxial test so you can see why it has the shape it does. The aim is that a reader can follow every line from the free energy down to the element tangent and reproduce the model in any FE code, not only this one.

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1 Scope and notation

We model fracture in concrete and rock with one scalar damage variable. That is a deliberate choice. A single ω cannot represent crack closure or direction-dependent stiffness loss, but for monotonic loading it is accurate, cheap, and easy to debug, and it is the workhorse of structural-scale fracture simulation. The whole formulation lives in small strains and rate independence.

Tensors are bold. Second-order tensors use a single bold letter ($\boldsymbol{\varepsilon}$, $\boldsymbol{\sigma}$), the fourth-order elastic tensor is $\mathbb{C}_0$, and $\mathbf{I}$ is the second-order identity. A colon is a double contraction, $\mathbb{C}_0 : \boldsymbol{\varepsilon}$ has components $C_{ijkl}\varepsilon_{kl}$, and a dyadic $\mathbf{a} \otimes \mathbf{b}$ has components $a_{ij}b_{kl}$. The strain invariants we need are

$$I_1 = \text{tr} \boldsymbol{\varepsilon} = \varepsilon_{kk}, \quad \mathbf{e} = \boldsymbol{\varepsilon} - \frac{1}{3} I_1 \mathbf{I}, \quad J_2 = \frac{1}{2} \mathbf{e} : \mathbf{e}, \quad (1)$$

where $\mathbf{e}$ is the deviatoric strain. The factor of one half in J_2 is not cosmetic; it is exactly what makes the equivalent strain reduce to the axial strain under uniaxial tension, and getting it wrong (a stray 1/6, say) shifts the whole damage threshold. We verify that in Section 3.

2 Damage kinematics and a short thermodynamic argument

2.1 Effective stress and the stiffness reduction

Start from the physical picture. A representative volume of material carries microcracks and voids. Let $\omega \in [0, 1]$ be the fraction of a cross-section that has been lost to those defects, so $\omega = 0$ is virgin material and $\omega = 1$ is fully broken. The load that used to pass through the whole section now passes through the remaining ligament, so the *effective* stress on the intact material is

$$\bar{\boldsymbol{\sigma}} = \frac{\boldsymbol{\sigma}}{1 - \omega}. \quad (2)$$

The intact material is still linear elastic, $\bar{\boldsymbol{\sigma}} = \mathbb{C}_0 : \boldsymbol{\varepsilon}$, which gives the nominal stress carried by the damaged solid:

$$\boxed{\boldsymbol{\sigma} = (1 - \omega) \mathbb{C}_0 : \boldsymbol{\varepsilon}} \quad (3)$$

This is the entire stress–strain law. Everything that follows is about one thing: how ω grows.

2.2 Why the law is allowed (the dissipation check)

It is worth confirming that (3) does not secretly create energy. Take the Helmholtz free energy per unit volume as the stored elastic energy scaled by the surviving stiffness,

$$\psi(\boldsymbol{\varepsilon}, \omega) = (1 - \omega) \frac{1}{2} \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} = (1 - \omega) \psi_0(\boldsymbol{\varepsilon}). \quad (4)$$

The Clausius–Duhem inequality for an isothermal process reads $\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} - \dot{\psi} \geq 0$. Differentiating ψ ,

$$\dot{\psi} = (1 - \omega) \mathbb{C}_0 : \boldsymbol{\varepsilon} : \dot{\boldsymbol{\varepsilon}} - \psi_0 \dot{\omega} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} - \psi_0 \dot{\omega}, \quad (5)$$

so the dissipation collapses to

$$\mathcal{D} = \psi_0(\boldsymbol{\varepsilon}) \dot{\omega} \geq 0. \quad (6)$$

Since $\psi_0 \geq 0$ always, the model is thermodynamically admissible as long as damage never decreases, $\dot{\omega} \geq 0$. That single constraint, no healing, is what the history variable in the next section enforces.

3 The modified von Mises equivalent strain

Damage should grow when the strain state is severe enough, but “severe” has to be a single number before we can compare it to a threshold. That number is the equivalent strain $\tilde{\varepsilon}$. Concrete is far weaker in tension than in compression, so $\tilde{\varepsilon}$ has to be asymmetric: a given tensile strain must count for much more than the same strain in compression. The modified von Mises measure of de Vree, Brekelmans and van Gils builds that asymmetry in through the strength ratio

$$k = \frac{f_c}{f_t}, \quad k \approx 10 \text{ for concrete.} \quad (7)$$

The measure is

$$\tilde{\varepsilon} = \underbrace{\frac{k-1}{2k(1-2\nu)} I_1}_{\text{volumetric, signed}} + \frac{1}{2k} \sqrt{\left(\frac{k-1}{1-2\nu}\right)^2 I_1^2 + \frac{12k}{(1+\nu)^2} J_2}. \quad (8)$$

The first term carries the sign of the volumetric strain, so tension and compression are treated differently from the start. The term under the root is positive either way and supplies the deviatoric (shape-change) contribution. The grouping of constants looks arbitrary until you push two load cases through it.

3.1 Why these exact constants: the uniaxial tension check

Put the material in uniaxial tension with axial strain $\varepsilon > 0$. The lateral strains are $-\nu\varepsilon$, so $\boldsymbol{\varepsilon} = \text{diag}(\varepsilon, -\nu\varepsilon, -\nu\varepsilon)$. The first invariant is

$$I_1 = \varepsilon(1-2\nu). \quad (9)$$

The deviatoric components come out to

$$e_{11} = \frac{2}{3}(1+\nu)\varepsilon, \quad e_{22} = e_{33} = -\frac{1}{3}(1+\nu)\varepsilon, \quad (10)$$

and the deviatoric invariant is

$$J_2 = \frac{1}{2}(e_{11}^2 + e_{22}^2 + e_{33}^2) = \frac{1}{2} \cdot \frac{6}{9}(1+\nu)^2 \varepsilon^2 = \frac{1}{3}(1+\nu)^2 \varepsilon^2. \quad (11)$$

Now feed these into the two pieces of (8). The deviatoric term inside the root cancels the $(1+\nu)^2$ cleanly,

$$\frac{12k}{(1+\nu)^2} J_2 = \frac{12k}{(1+\nu)^2} \cdot \frac{1}{3}(1+\nu)^2 \varepsilon^2 = 4k \varepsilon^2, \quad (12)$$

and the volumetric term inside the root cancels the $(1-2\nu)$,

$$\left(\frac{k-1}{1-2\nu}\right)^2 I_1^2 = \left(\frac{k-1}{1-2\nu}\right)^2 \varepsilon^2 (1-2\nu)^2 = (k-1)^2 \varepsilon^2. \quad (13)$$

So the root simplifies to a perfect square:

$$\sqrt{(k-1)^2 \varepsilon^2 + 4k \varepsilon^2} = \varepsilon \sqrt{k^2 + 2k + 1} = \varepsilon(k+1). \quad (14)$$

The leading term is $\frac{k-1}{2k(1-2\nu)} \varepsilon(1-2\nu) = \frac{k-1}{2k} \varepsilon$, and putting both together,

$$\tilde{\varepsilon} = \frac{k-1}{2k} \varepsilon + \frac{1}{2k} \varepsilon(k+1) = \frac{\varepsilon}{2k} [(k-1) + (k+1)] = \varepsilon. \quad (15)$$

That is the point of the constants. Under uniaxial tension the equivalent strain is just the axial strain, so the threshold κ_0 is read straight off a tension test with no correction factor.

3.2 And the compression check

Repeat with uniaxial compression, axial strain $-\varepsilon$ ($\varepsilon > 0$). Now $I_1 = -\varepsilon(1 - 2\nu)$, the first term flips sign, the root is unchanged, and

$$\tilde{\varepsilon} = -\frac{k-1}{2k}\varepsilon + \frac{k+1}{2k}\varepsilon = \frac{\varepsilon}{2k}[-(k-1) + (k+1)] = \frac{\varepsilon}{k}. \quad (16)$$

Compression reaches the same $\tilde{\varepsilon}$ only at k times the strain. Damage therefore starts in compression at roughly k times the tensile load, which is exactly the strength ratio f_c/f_t we asked for. The measure does what concrete does.

4 Damage growth: loading function and history

Damage tracks the worst strain the point has ever felt. Define a history variable κ and a loading function

$$f(\tilde{\varepsilon}, \kappa) = \tilde{\varepsilon} - \kappa. \quad (17)$$

The state must satisfy the Kuhn–Tucker (loading–unloading) conditions

$$f \leq 0, \quad \dot{\kappa} \geq 0, \quad \dot{\kappa} f = 0. \quad (18)$$

Read them in plain terms. The strain state can never sit outside the current envelope ($f \leq 0$); the memory only ever grows ($\dot{\kappa} \geq 0$); and it grows only while the state is pushing on the envelope ($\dot{\kappa} f = 0$). Integrated over a monotonic-in-memory history this is simply

$$\kappa(t) = \max\left(\kappa_0, \max_{\tau \leq t} \tilde{\varepsilon}(\tau)\right), \quad (19)$$

with the initial threshold

$$\kappa_0 = \frac{f_t}{E}. \quad (20)$$

Below κ_0 the material is elastic and $\omega = 0$. The instant $\tilde{\varepsilon}$ first exceeds κ_0 , damage starts and never reverses, which is the no-healing condition the dissipation argument demanded.

5 Exponential softening

We need a rule $\omega(\kappa)$ that climbs from 0 at κ_0 toward 1 as κ grows. FRACMAT uses the exponential law

$$\omega(\kappa) = 1 - \frac{\kappa_0}{\kappa} \exp\left[-\frac{\kappa - \kappa_0}{\varepsilon_f - \kappa_0}\right], \quad \kappa \geq \kappa_0. \quad (21)$$

Two limits check out by inspection: at $\kappa = \kappa_0$ the exponential is 1 and the prefactor is 1, so $\omega = 0$; as $\kappa \rightarrow \infty$ both the κ_0/κ factor and the exponential go to zero, so $\omega \rightarrow 1$. The parameter ε_f sets how fast the material lets go, and we will pin it down from fracture energy in the next section.

A detail that matters for the solver: the uniaxial stress during monotonic loading takes a clean closed form. With $\varepsilon = \kappa$ during loading and $E\kappa_0 = f_t$,

$$\sigma = (1 - \omega)E\kappa = E\kappa_0 \exp\left[-\frac{\kappa - \kappa_0}{\varepsilon_f - \kappa_0}\right] = f_t \exp\left[-\frac{\kappa - \kappa_0}{\varepsilon_f - \kappa_0}\right]. \quad (22)$$

The stress decays exponentially from the tensile strength. We use this in a moment to get the dissipated energy in one integral.

We will also need the slope of the damage law for the tangent. Differentiating (21) with the product rule,

$$\frac{d\omega}{d\kappa} = \frac{\kappa_0}{\kappa} \exp\left[-\frac{\kappa - \kappa_0}{\varepsilon_f - \kappa_0}\right] \left(\frac{1}{\kappa} + \frac{1}{\varepsilon_f - \kappa_0}\right), \quad \kappa \geq \kappa_0. \quad (23)$$

It is positive throughout, as a softening law must be, since every factor is positive.

6 Crack band regularization and the calibration of ε_f

Plain softening has a famous problem. Refine the mesh and the damage localizes into one element, the dissipated energy shrinks with the element size, and the answer never converges. The crack band method fixes this by making the local softening branch depend on the element size so that the energy released per unit crack *area* stays fixed at the material fracture energy G_F .

6.1 Energy under the softening curve

The energy dissipated per unit volume in a fully softened material point is the area under the uniaxial stress–strain curve. Split it into the elastic triangle up to κ_0 and the softening tail from (22):

$$g_f = \underbrace{\frac{1}{2}f_t\kappa_0}_{\text{elastic}} + \underbrace{\int_{\kappa_0}^{\infty} f_t \exp\left[-\frac{\kappa - \kappa_0}{\varepsilon_f - \kappa_0}\right] d\kappa}_{\text{softening tail}} \quad (24)$$

$$= \frac{1}{2}f_t\kappa_0 + f_t(\varepsilon_f - \kappa_0) = f_t\left(\varepsilon_f - \frac{1}{2}\kappa_0\right). \quad (25)$$

The tail integral is elementary: the integral of a decaying exponential is just its decay length $(\varepsilon_f - \kappa_0)$ times the starting value f_t .

6.2 Tie the volume energy to the area energy

Crack band regularization says the energy per unit *area*, G_F , equals the energy per unit *volume*, g_f , smeared over a band one characteristic length h wide:

$$G_F = h g_f. \quad (26)$$

Substitute (25) and solve for the only free parameter:

$$\boxed{\varepsilon_f = \frac{G_F}{h f_t} + \frac{\kappa_0}{2}} \quad (27)$$

Each element gets its own ε_f from its own h . A big element softens faster (smaller ε_f), a small element softens slower, and the product $h g_f$ comes out to G_F either way. That is what makes the peak load and the total dissipation independent of the mesh.

6.3 The characteristic length: Oliver’s projection

The remaining question is what h to use. For a linear triangle, FRACMAT uses Oliver’s projection,

$$h = \frac{2A_e}{\ell_{\max}}, \quad (28)$$

where A_e is the element area and $\ell_{\max}$ its longest edge. Geometrically this is the width of the element measured across the band, i.e. the height of the triangle on its longest side. The longest edge is taken as the proxy for the crack direction, so h is the distance the crack has to traverse. For 3D elements the same idea uses the element volume and the projected area of the dominant face. The practical payoff, and the reason the torsion benchmark works at all, is that h follows the local crack orientation instead of being locked to a fixed mesh diagonal.

7 From the continuum to a discrete residual

The equilibrium statement is the principle of virtual work: for every admissible virtual displacement $\delta \mathbf{u}$,

$$\int_{\Omega} \delta \boldsymbol{\varepsilon} : \boldsymbol{\sigma} \, dV = \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{b} \, dV + \int_{\Gamma_t} \delta \mathbf{u} \cdot \mathbf{t} \, dS. \quad (29)$$

Interpolate the displacement with shape functions, $\mathbf{u} = \mathbf{N} \mathbf{d}$, so that strain is $\boldsymbol{\varepsilon} = \mathbf{B} \mathbf{d}$ with $\mathbf{B}$ the usual strain–displacement matrix. The virtual work statement turns into a set of nonlinear algebraic equations in the nodal displacements $\mathbf{d}$:

$$\mathbf{r}(\mathbf{d}) = \mathbf{f}^{\text{int}}(\mathbf{d}) - \mathbf{f}^{\text{ext}} = \mathbf{0}, \quad \mathbf{f}^{\text{int}} = \int_{\Omega} \mathbf{B}^T \boldsymbol{\sigma} \, dV. \quad (30)$$

The internal force is nonlinear in $\mathbf{d}$ because $\boldsymbol{\sigma}$ depends on ω , which depends on the strain through $\tilde{\varepsilon}$ and the history. We solve (30) under displacement control, prescribing the loaded degrees of freedom and iterating on the free ones.

8 The Newton–Raphson solve

8.1 The consistent tangent, derived in full

Newton’s method linearizes the residual: given an iterate $\mathbf{d}^{(i)}$ with residual $\mathbf{r}^{(i)}$, solve $\mathbf{K}_T \delta \mathbf{d} = -\mathbf{r}^{(i)}$ and update $\mathbf{d}^{(i+1)} = \mathbf{d}^{(i)} + \delta \mathbf{d}$. The tangent is $\mathbf{K}_T = \partial \mathbf{f}^{\text{int}} / \partial \mathbf{d} = \int_{\Omega} \mathbf{B}^T \mathbf{D}_T \mathbf{B} \, dV$, so the whole problem reduces to the material tangent $\mathbf{D}_T = d\boldsymbol{\sigma} / d\boldsymbol{\varepsilon}$.

Differentiate the stress law (3). During active loading the history follows the equivalent strain, $\kappa = \tilde{\varepsilon}$ and $d\kappa = d\tilde{\varepsilon}$, so by the product rule

$$d\boldsymbol{\sigma} = (1 - \omega) \mathbb{C}_0 : d\boldsymbol{\varepsilon} - \frac{d\omega}{d\kappa} (\mathbb{C}_0 : \boldsymbol{\varepsilon}) d\tilde{\varepsilon}. \quad (31)$$

The first term is the secant stiffness acting on the strain increment. The second is the loss of stiffness as damage advances. Writing $d\tilde{\varepsilon} = \frac{\partial \tilde{\varepsilon}}{\partial \boldsymbol{\varepsilon}} : d\boldsymbol{\varepsilon}$, the consistent tangent is

$$\boxed{\mathbf{D}_T = (1 - \omega) \mathbb{C}_0 - \frac{d\omega}{d\kappa} (\mathbb{C}_0 : \boldsymbol{\varepsilon}) \otimes \frac{\partial \tilde{\varepsilon}}{\partial \boldsymbol{\varepsilon}}} \quad (32)$$

The first part is symmetric. The second is a rank-one dyad and is generally not symmetric, which is the source of the unsymmetric, sometimes ill-conditioned tangents that make full Newton stall on the steep part of the softening branch.

8.2 The gradient of the equivalent strain

The dyad in (32) needs $\partial \tilde{\varepsilon} / \partial \boldsymbol{\varepsilon}$. Write (8) compactly with

$$a = \frac{k - 1}{2k(1 - 2\nu)}, \quad b = \frac{k - 1}{1 - 2\nu}, \quad c = \frac{12k}{(1 + \nu)^2}, \quad S = \sqrt{b^2 I_1^2 + c J_2}, \quad (33)$$

so that $\tilde{\varepsilon} = a I_1 + \frac{1}{2k} S$. Using $\partial I_1 / \partial \boldsymbol{\varepsilon} = \mathbf{I}$ and $\partial J_2 / \partial \boldsymbol{\varepsilon} = \mathbf{e}$ (which follows from $J_2 = \frac{1}{2} \mathbf{e} : \mathbf{e}$ and the deviatoric projection),

$$\frac{\partial \tilde{\varepsilon}}{\partial \boldsymbol{\varepsilon}} = a \mathbf{I} + \frac{1}{2k} \cdot \frac{1}{2S} (2b^2 I_1 \mathbf{I} + c \mathbf{e}) = a \mathbf{I} + \frac{b^2 I_1}{2kS} \mathbf{I} + \frac{c}{4kS} \mathbf{e}. \quad (34)$$

As $S \rightarrow 0$ near a stress-free state this expression is singular, which is harmless because damage is inactive there and the dyad term is never assembled; the code guards the divide with the loading check anyway.

8.3 What FRACMAT actually solves: the modified (secant) scheme

Full Newton with (32) converges quadratically but is fragile once the dyad term turns the tangent unsymmetric and nearly singular. FRACMAT instead uses the *modified* Newton–Raphson scheme: keep only the symmetric secant part,

$$\mathbf{D}_T \approx (1 - \omega) \mathbb{C}_0, \quad \mathbf{K}_T = \int_{\Omega} \mathbf{B}^T (1 - \omega) \mathbb{C}_0 \mathbf{B} \, dV. \quad (35)$$

This drops the rank-one term entirely. The trade is deliberate: convergence slows from quadratic to linear, but the tangent stays symmetric positive definite, the factorization is stable, and it can be reused across iterations within a step. The residual (30) still uses the true damaged stress, so the converged solution is the exact one; only the path to it is slower. On the benchmarks this buys robustness through the snap-back-prone softening regime at a cost of a few extra iterations per step, which the displacement-controlled stepping absorbs.

8.4 Solver loop

Modified Newton–Raphson, one displacement-controlled step

1. Apply the prescribed displacement increment to the loaded DOFs; freeze the supports.
2. **For** iteration $i = 0, 1, 2, \dots$ until $\|\mathbf{r}\| \leq \text{tol}$:
 - (a) For every element, compute $\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{d}$, then I_1 , J_2 , and $\tilde{\varepsilon}$ from (8).
 - (b) Update history $\kappa = \max(\kappa, \tilde{\varepsilon})$ from (19); if $\kappa > \kappa_0$, update ω from (21) with the element’s ε_f from (27).
 - (c) Assemble $\boldsymbol{\sigma} = (1 - \omega)\mathbb{C}_0 : \boldsymbol{\varepsilon}$ and the internal force $\mathbf{f}^{\text{int}} = \int \mathbf{B}^T \boldsymbol{\sigma} \, dV$.
 - (d) Form the residual $\mathbf{r} = \mathbf{f}^{\text{int}} - \mathbf{f}^{\text{ext}}$ on the free DOFs.
 - (e) Assemble the secant tangent $\mathbf{K}_T$ from (35); solve $\mathbf{K}_T \delta \mathbf{d} = -\mathbf{r}$ with sparse backslash (UMFPACK).
 - (f) Update $\mathbf{d} \leftarrow \mathbf{d} + \delta \mathbf{d}$.
3. Commit the converged κ and ω as the history for the next step.

All of step 2(a)–(c) runs over every element in one batched tensor operation (`pagetimes` in MATLAB), which is where the vectorized speedup comes from. The sparse solve in 2(e) dominates the wall-clock time, as the timing table in the JOSS paper shows.

9 Symbol reference

Symbol	Meaning	Symbol	Meaning
ω	scalar damage, $[0, 1]$	E, ν	Young’s modulus, Poisson ratio
$\boldsymbol{\sigma}, \boldsymbol{\varepsilon}$	stress, strain	$\mathbb{C}_0$	virgin elastic stiffness
$\tilde{\varepsilon}$	equivalent strain	κ	history variable
I_1	first strain invariant	J_2	second deviatoric invariant
$\boldsymbol{e}$	deviatoric strain	$k = f_c/f_t$	strength ratio
f_t, f_c	tensile, compressive strength	$\kappa_0 = f_t/E$	damage onset threshold
ε_f	softening parameter	G_F	fracture energy
g_f	energy per unit volume	h	characteristic length
$A_e, \ell_{\max}$	element area, longest edge	$\boldsymbol{B}$	strain–displacement matrix
$\boldsymbol{r}$	residual	$\boldsymbol{K}_T$	tangent stiffness
$\boldsymbol{D}_T$	material tangent	$\boldsymbol{d}$	nodal displacements

Primary references. de Vree, Brekelmans & van Gils (1995) for the equivalent strain; Bažant & Oh (1983) for crack band theory; Oliver (1989) for the characteristic-length projection; Peerlings et al. (1996) and Jirásek (2004) for the damage formulation and its numerics. Full citations are in the repository `paper.bib`.